

MSc Project Plan

CSC-44120

Project Overview and Description

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Degree Title: Artificial Intelligence and Data Science (Master of Science)

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Project Title: Self-Reflective Retrieval-Augmented Generation: Improving Attribution via Iterative Verification

Brief Project Description

Large Language Models (LLMs) frequently generate responses that appear convincing but may contain hallucinations or unsupported statements. Traditional Retrieval-Augmented Generation (RAG) systems attempt to reduce hallucinations by retrieving relevant documents before generating an answer. However, standard RAG typically performs retrieval only once before generating a response, meaning errors in retrieval or reasoning are propagated directly into the final answer.

This project aims to develop a Self-Reflective RAG framework that introduces an iterative verification mechanism. After an initial answer is generated, an LLM-based verifier evaluates whether each claim is supported by retrieved evidence. If unsupported or inconsistent information is detected, additional retrieval is performed and the answer is refined until satisfactory attribution and factual consistency are achieved.

The system will be evaluated using benchmark datasets including HotpotQA and FEVER. Performance will be compared against a traditional single-pass RAG system using multiple evaluation metrics including retrieval accuracy, attribution quality, factual correctness and hallucination rate.

Aims and Objectives of the Project

Aim: To design, implement and evaluate a self-reflective Retrieval-Augmented Generation system that improves factual accuracy and source attribution through iterative verification.

Objectives:

- Study recent research on Retrieval-Augmented Generation and hallucination mitigation.
- Implement a baseline single-pass RAG pipeline.

- Build a multi-stage pipeline consisting of: Document Retrieval, Answer Generation, Verification, and Refinement.
- Develop an LLM-based verifier capable of detecting unsupported claims.
- Implement iterative retrieval and answer refinement.
- Evaluate performance using HotpotQA and FEVER datasets.
- Compare the proposed system against conventional RAG.
- Analyse improvements in attribution, factual consistency and hallucination reduction.

Brief overview of the key literature related to the Project

Retrieval-Augmented Generation (Lewis et al., 2020) combines external knowledge retrieval with language generation to improve factual grounding. While RAG reduces hallucinations, recent studies demonstrate that retrieval alone cannot eliminate unsupported claims because the generated response may still misinterpret retrieved evidence.

Recent work including Self-RAG (Asai et al., 2024), CRAG (Yan et al., 2024), Reflexion (Shinn et al., 2023) and Corrective RAG introduces self-evaluation and iterative refinement to improve response quality. These systems enable language models to assess their own outputs and perform additional retrieval when evidence is insufficient.

Research on benchmark datasets such as HotpotQA and FEVER provides reliable evaluation frameworks for multi-hop reasoning and fact verification respectively. These datasets are widely adopted for evaluating retrieval quality, factual correctness and attribution performance. This project builds upon these advances by implementing an iterative retrieve–generate–verify–refine pipeline and comparing it with a conventional RAG baseline.

Project Processes and Methods

Methodology

The project will follow an experimental software engineering methodology. A baseline RAG system will first be implemented using a vector database and dense retrieval. A second system will extend this baseline by introducing an iterative verification stage where an LLM evaluates factual consistency between generated responses and retrieved evidence. When inconsistencies are identified, the system performs additional retrieval before regenerating the response. Both systems will be evaluated using identical datasets and quantitative metrics.

Data Collection Methods

No primary data collection involving human participants will be undertaken. Public benchmark datasets including HotpotQA and FEVER will be used for experimentation and evaluation.

Skills Analysis

Skills already acquired:

- Python programming
- Machine Learning & Deep Learning
- Natural Language Processing
- Information Retrieval
- Data Analysis
- PyTorch & Transformers
- Git
- Research methodology

Skills to develop:

- Advanced Retrieval-Augmented Generation
 - FAISS vector databases
 - LangChain / LlamaIndex
 - HuggingFace ecosystem
 - LLM evaluation techniques
 - Prompt engineering
 - Experiment design & Statistical evaluation
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Time and Resource Planning

Will Standard Departmental Hardware be used? YES

Will Software which is already available in department be used? YES

- **Software list:** Python, VS Code, Jupyter Notebook, PyTorch, HuggingFace Transformers, Sentence Transformers, LangChain, FAISS, ChromaDB, Pandas, NumPy, Scikit-learn, Matplotlib, Git. (All are open-source).

Will the project require any Programming? YES

- **Programming Languages:** Python
- **IDEs and Libraries:** VS Code, Jupyter Notebook; Transformers, Torch, LangChain, FAISS, ChromaDB, SentenceTransformers, Pandas, NumPy, Scikit-learn, Evaluate, NLTK.

Table of Risks

Risk-id	Description	Probability	Best practice prevention measures	Remedy
R1	Large model requires high GPU memory	Medium	Use smaller open-source models	Use Google Colab Pro
R2	Low retrieval quality	Medium	Tune embeddings and retrieval parameters	Experiment with different embedding models
R3	Long execution time	Medium	Cache retrieval results	Reduce dataset size
R4	Library compatibility	Low	Use virtual environments	Update package versions
R5	Limited evaluation time	Medium	Prepare milestones early	Reduce experiment scope

Gantt Chart (June - September 2026)

Task / Activity	Jun	Jul	Aug	Sep
Project Allocation & Supervisor Meeting				
Literature Review				
Requirements Analysis & System Design				
Project Plan & Ethics Form Submission (08 Jul)				
Baseline RAG Implementation				
Retriever & Vector Database Development				
Generator Module Development				
LLM-based Verification Module				
Iterative Retrieval & Refinement Pipeline				

Task / Activity	Jun	Jul	Aug	Sep
Poster Preparation		■		
Poster Submission (29 Jul)		■		
Experimental Evaluation (HotpotQA & FEVER)			■■■	
Performance Comparison (Baseline vs RAG)			■■■	
Results Analysis			■■■	■
Dissertation Writing		■	■■■	■■■
Poster Presentation (19 Aug)			■	
Final Editing & Proofreading				■■■
Final Report Submission (03 Sep)				■

Milestones

- **15 June 2026:** Supervisor allocated and first meeting
- **08 July 2026:** Project Plan & GDPR/Ethics submission
- **29 July 2026:** Poster submission
- **19 August 2026:** Poster presentation
- **03 September 2026:** Final dissertation submission
- **September 2026:** Project demonstration

References

- Asai, A. et al. (2024). *Self-RAG: Learning to Retrieve, Generate and Critique through Self-Reflection.*
- Lewis, P. et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.*

- Shinn, N. et al. (2023). *Reflexion: Language Agents with Verbal Reinforcement Learning*.
- Thorne, J. et al. (2018). *FEVER: A Large-scale Dataset for Fact Extraction and Verification*.
- Yan, S. et al. (2024). *Corrective Retrieval-Augmented Generation*.
- Yang, Z. et al. (2018). *HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering*.

Submission Date: July 07, 2026